CHAPTER 6:

Vacuum Decay

> None of the things one frets about ever happen. Something one's never thought of does.
>
> Connie Willis, *Doomsday Book*

In March 2008, a retired nuclear safety officer named Walter Wagner filed a lawsuit against the U.S. government to prevent scientists from starting up the Large Hadron Collider. From Wagner's point of view, it was a desperate bid to save the world. The lawsuit was, of course, doomed to fail. For one thing, the LHC is controlled by the European Organization for Nuclear Research (abbreviated as CERN; the acronym comes from the French), not the U.S. government. And Wagner's scientific worries, though presumably sincerely felt, were unfounded. In the end, CERN leadership put out some reassuring press releases about the safety of its collider technology, and the construction and operation of the LHC continued.

That didn't stop some segments of the public from ramping up their panic as the date of the first scheduled particle collisions approached. The LHC would be the most powerful particle physics experiment in history, colliding protons in four places along a giant circular, supercooled, vacuum-sealed underground track 27 kilometers in circumference. These collisions would produce, within the detectors, momentary bursts of energy so powerful they could re-create the conditions of the Hot Big Bang mere nanoseconds after the first moment of creation. The LHC would, the scientists hoped, lend us insight not only into the conditions of the early universe, but into the very structure of matter and energy itself. Earlier experiments had shown us that the laws of physics are energy-dependent—altering how particles and forces interact depending on the conditions in which they're found—so creating collisions of higher and higher energies would allow scientists to probe the edges of our understanding of how physics works.

And there was an even more tantalizing prize in sight. Decades before, physicists had theorized the existence of a new particle—one so central to the behavior of matter that it would be the final piece completing the